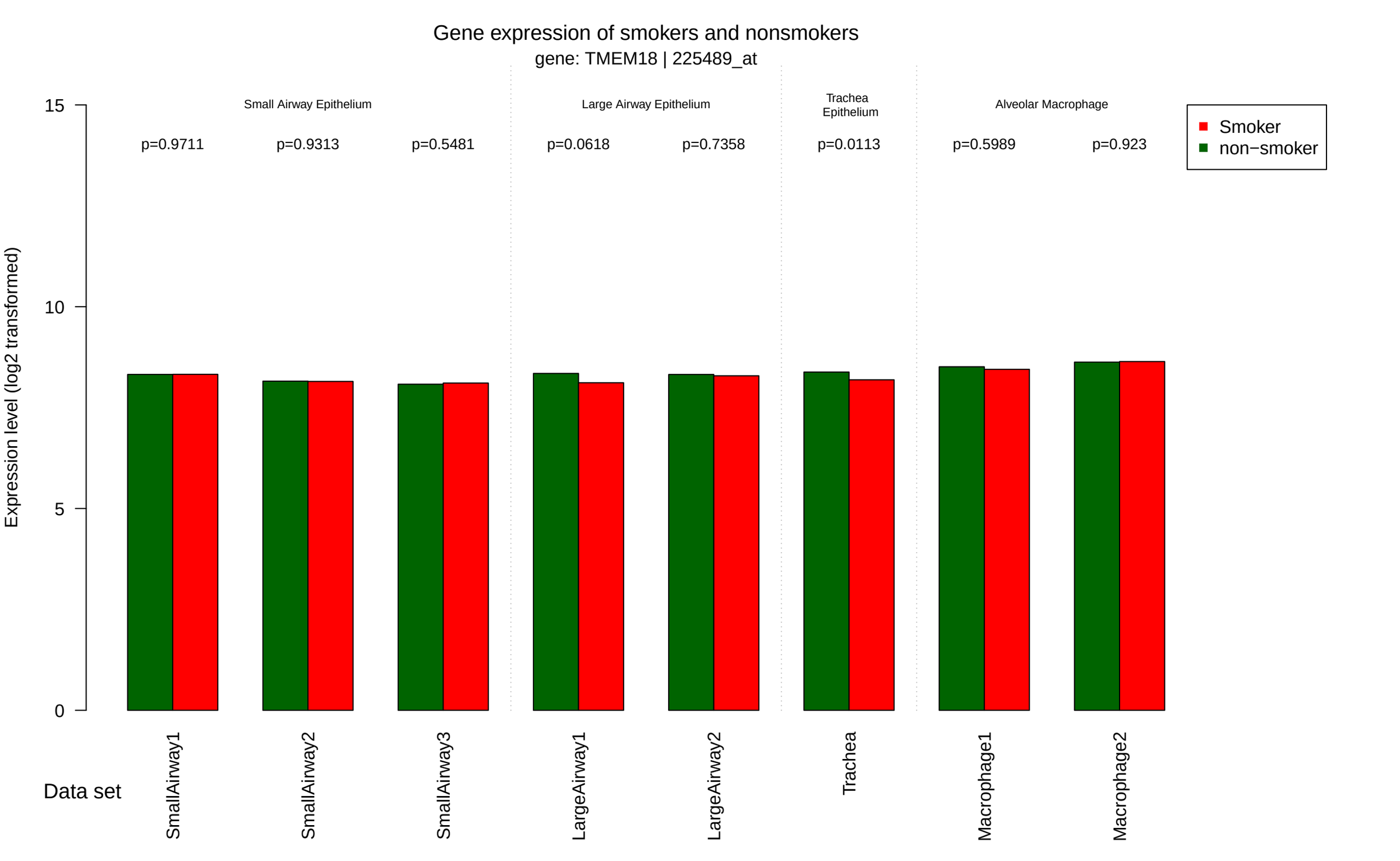
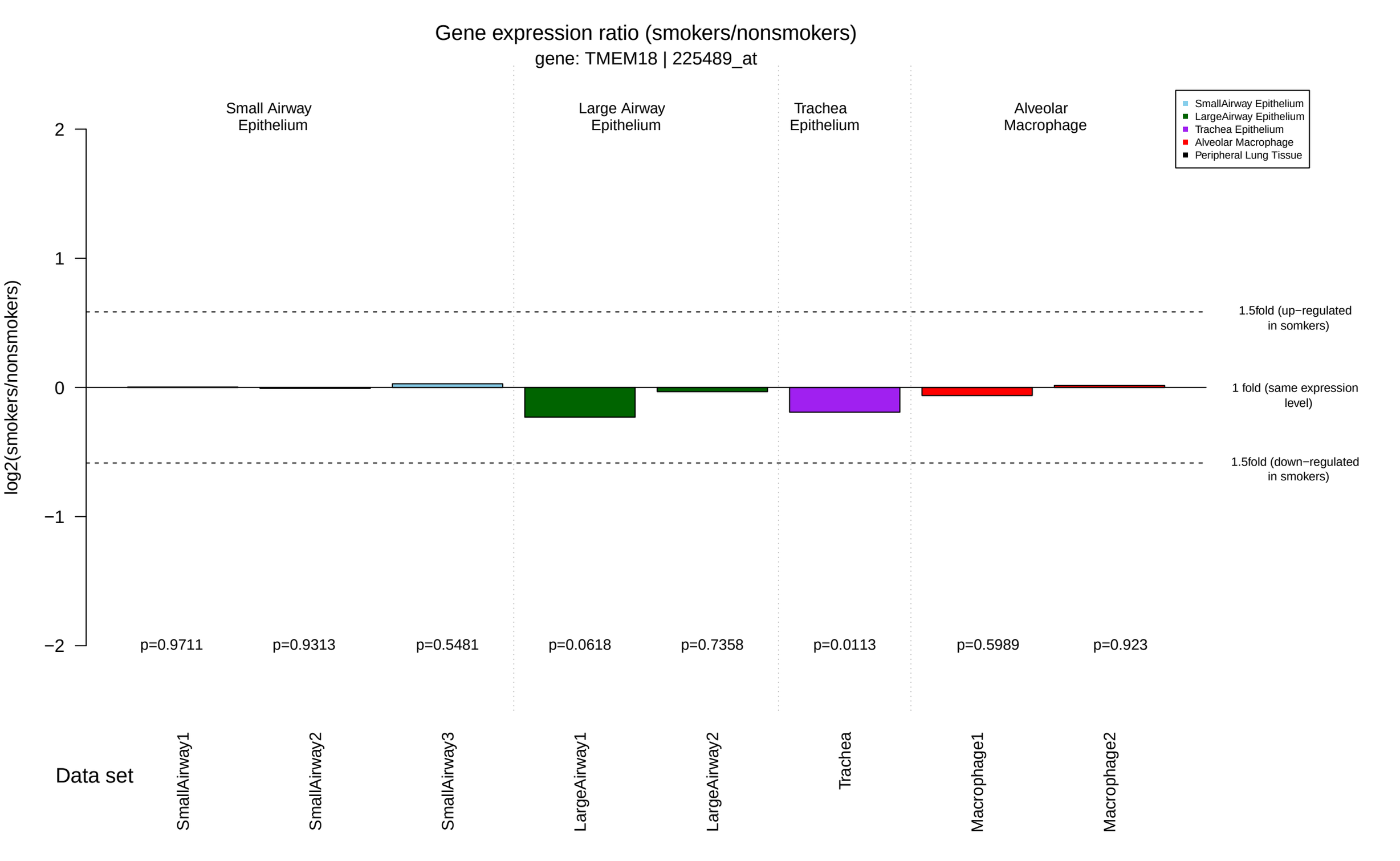
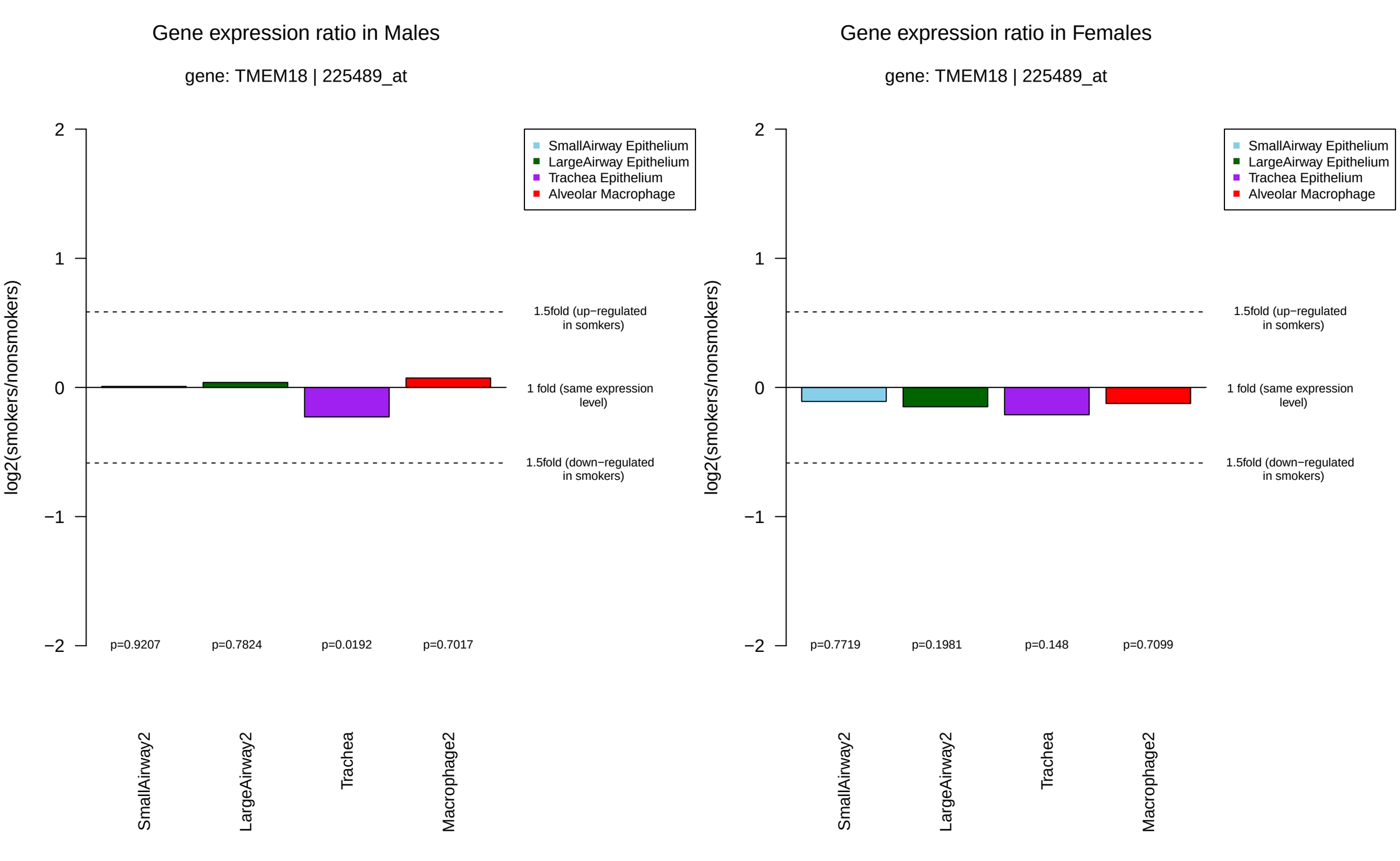
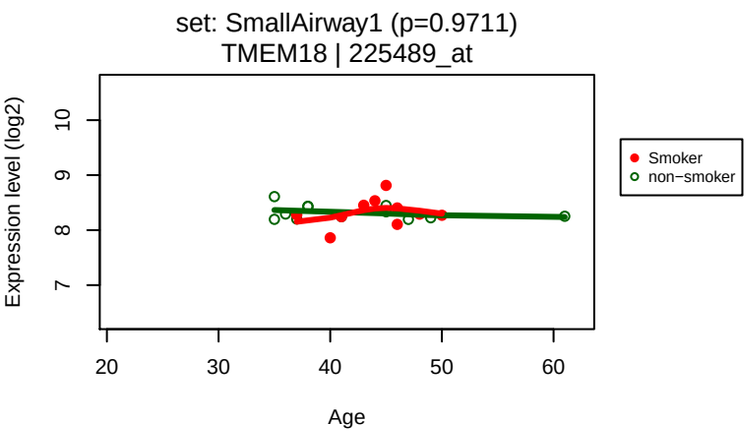
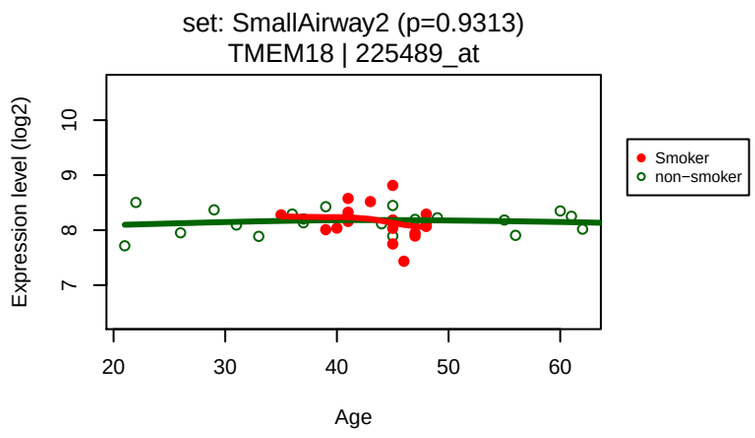
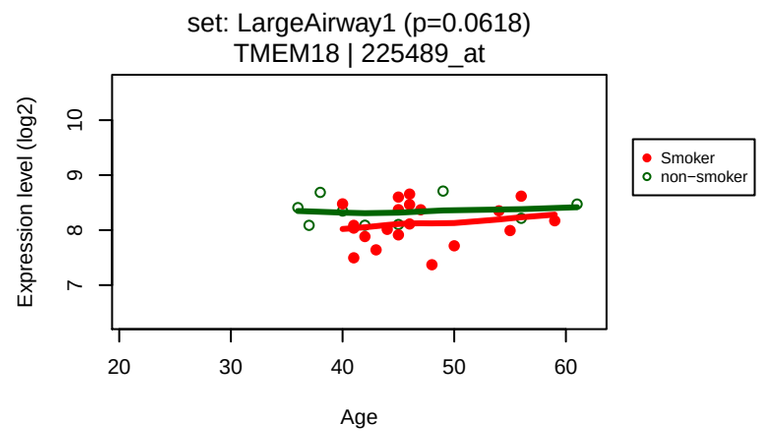

Gene expression of smokers and nonsmokers
gene: TMEM18 | 225489_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9711
p=0.9313
p=0.5481
p=0.0618
p=0.7358
p=0.0113
p=0.5989
p=0.923
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

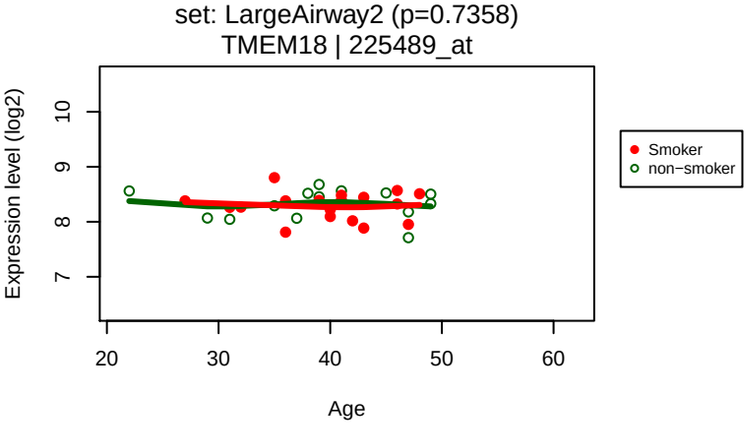

Gene expression ratio (smokers/nonsmokers)
gene: TMEM18 | 225489_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9711
p=0.9313
p=0.5481
p=0.0618
p=0.7358
p=0.0113
p=0.5989
p=0.923
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

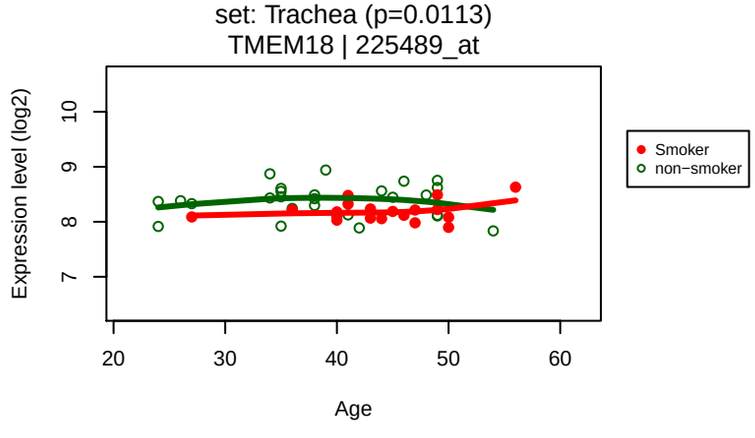

Gene expression ratio in Males
gene: TMEM18 | 225489_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9207
p=0.7824
p=0.0192
p=0.7017
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM18 | 225489_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7719
p=0.1981
p=0.148
p=0.7099
SmallAirway2
LargeAirway2
Trachea
Macrophage2

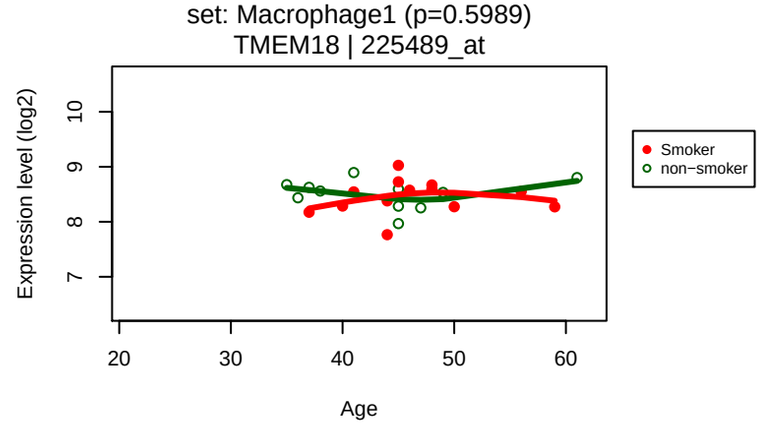

set: SmallAirway1 (p=0.9711)
TMEM18 | 225489_at
Expression level (log2)
Age
Smoker
non−smoker

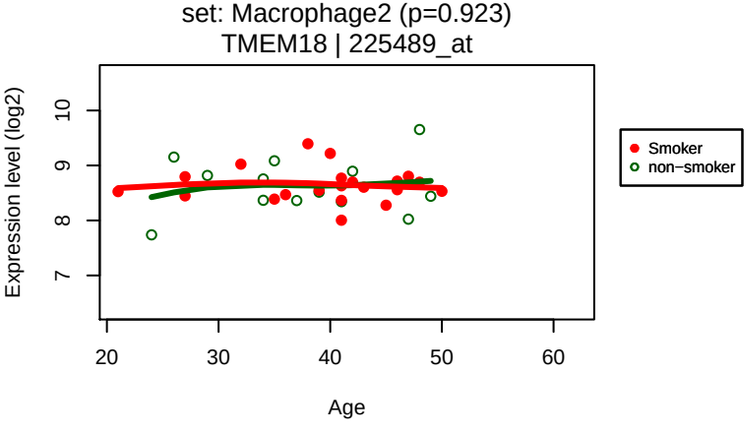

set: SmallAirway2 (p=0.9313)
TMEM18 | 225489_at
Expression level (log2)
Age
Smoker
non−smoker

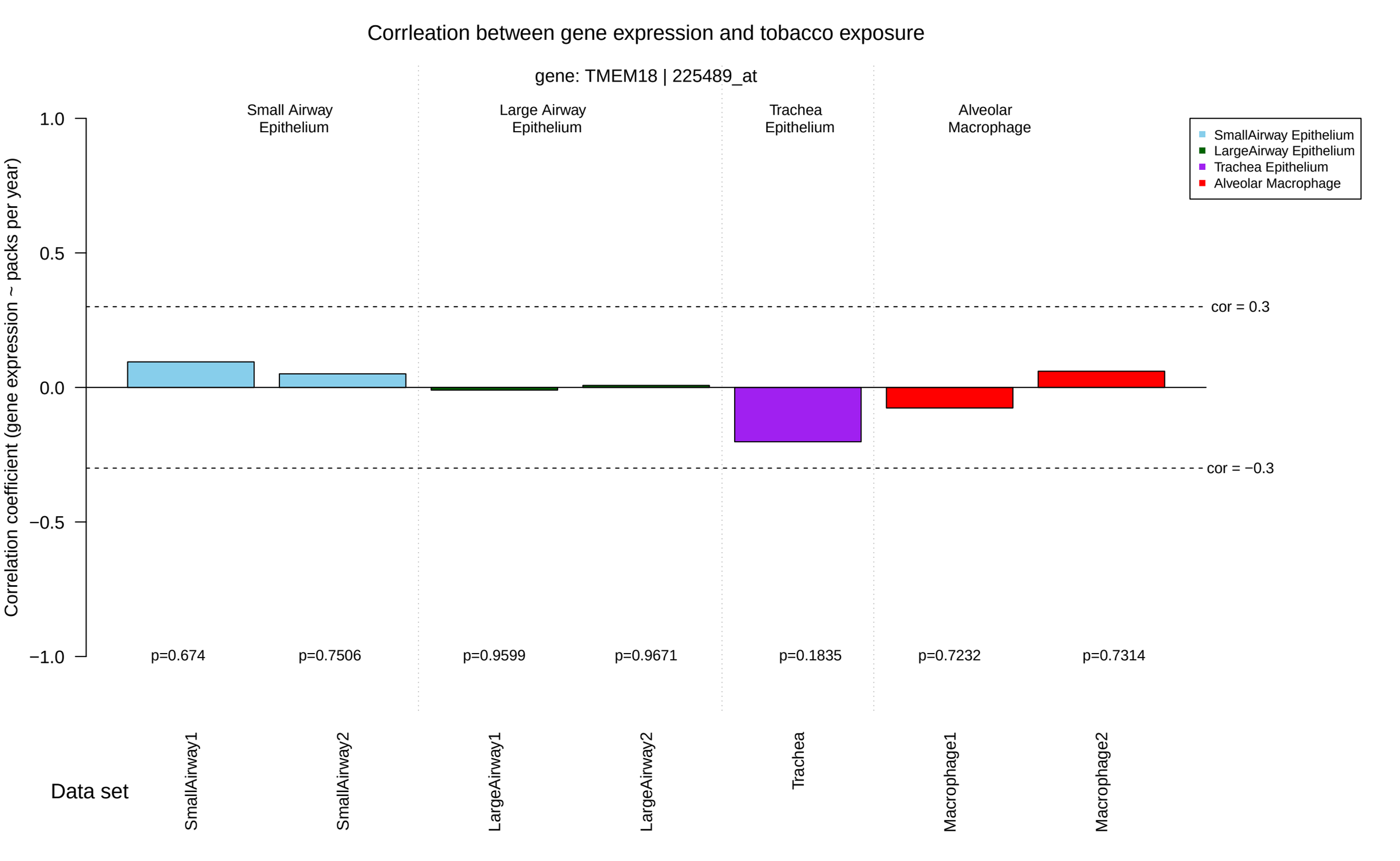

set: LargeAirway1 (p=0.0618)
TMEM18 | 225489_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7358)
TMEM18 | 225489_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0113)
TMEM18 | 225489_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5989)
TMEM18 | 225489_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.923)
TMEM18 | 225489_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM18 | 225489_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.674
p=0.7506
p=0.9599
p=0.9671
p=0.1835
p=0.7232
p=0.7314
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2